

Chap 8 - Elements of Group 1 & 2

Group 1 + Group 2 = s-block element

Group-1 $\rightarrow$ Hydrogen + alkali metals

Group-2 $\rightarrow$ alkaline Earth metal.

* Hydrogen Atom (or) dihydrogen

1) Hydrogen is the 1st element of the periodic table.

2) Atomic number of Hydrogen = $Z = 1$

3) no. of protons = 1

no. of electron = 1

3) Hydrogenation exist in Diatomic (H_2) form called as Dihydrogen

* occurrence

1) Dihydrogen is most abundant element in the earth.

2) Dihydrogen shows 70% total abundance.

3) Abundance of Dihydrogen $\rightarrow$ on mass bases $\rightarrow$ 10th most abundant element
on atom basis $\rightarrow$ 3rd most abundant element.

* Position of hydrogen in Periodic table.

1) Electronic (general) Configuration $\rightarrow$

Group 1 elements $\rightarrow$ alkali metal

General electronic configuration $\rightarrow ns^1$

$\therefore H = ns^1 \xrightarrow{+1e^-} ns^2$

G-17 $= ns^2np^5 \xrightarrow{+1e^-} ns^2np^6$

Hydrogen shows similar properties / resemblance as that of G-17 element & Group-1 element

2) Uniqueness of Hydrogen atom

$H \rightarrow H^+ + 1e^-$

Hydrogen atom proton electron

$H^+ \rightarrow$ Proton $\rightarrow$ Unstable

$\therefore H^+ + H_2O \rightarrow H_3O^+$ Hydronium ion (Stable)

$\rightarrow H^+$ (proton) as it is not stable, so that associates with H_2O (water) to form H_3O^+ ion

* Isotopes of Hydrogen.

Element having same atomic number but having different atomic mass are called as Isotope.

Atom representation $A \leftarrow$ Atomic mass

atomic number $\rightarrow Z$

Ex.



There are 3 isotopes of Hydrogen.

	Hydrogen	Deuterium	Tritium
Representation	${}^1_1\text{H}$	${}^2_1\text{H}$	${}^3_1\text{H}$
Atomic No.	1	1	1
No. of proton	1	1	1
No. of electron	1	1	1
No. of neutron	$(1-1)=$	$(2-1)=$	$(3-1)=$
$n = A - Z$	0 neutron	1 neutron	2 neutron
Diagram			

* preparation of dihydrogen (H_2)

Laboratory method

by reaction with acid

by reaction with base

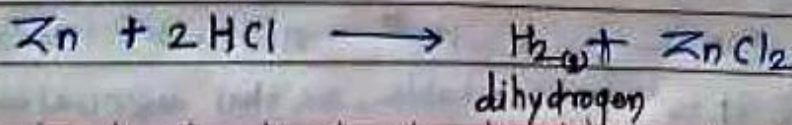
Industrial method

By electrolysis of pure H_2O

By form Carbon or hydrocarbon

① Laboratory method

② By reaction with acid (HCl)

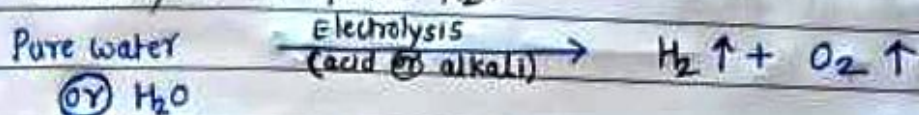


⑥ By reaction with base.



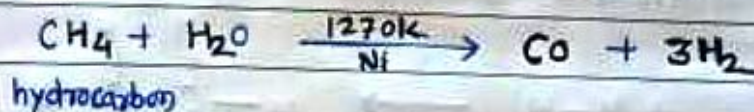
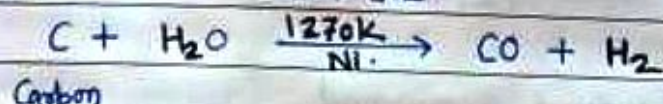
② Industrial method.

① electrolysis of pure H_2O

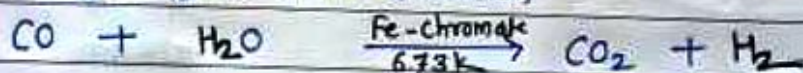


⑥ By Carbon (or) hydrocarbon.

step I - Formation of water Gas.



step II Water gas shift reaction



step-III Removal of CO_2

CO_2 formed is removed by sodium Arsenite.

* Chemical properties of dihydrogen (H_2).

1) Reaction ~~with~~ of dihydrogen with alkali metals.



② Reaction of dihydrogen with oxygen (O_2)



3) Reaction of dihydrogen with halogen



* Reducing property of dihydrogen

1) Reduction of oxide.

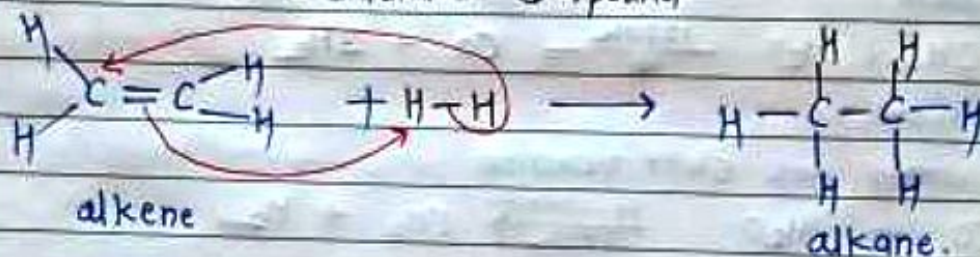


Cuprous oxide

Removal of oxygen $\longrightarrow$ Reduction Reaction.



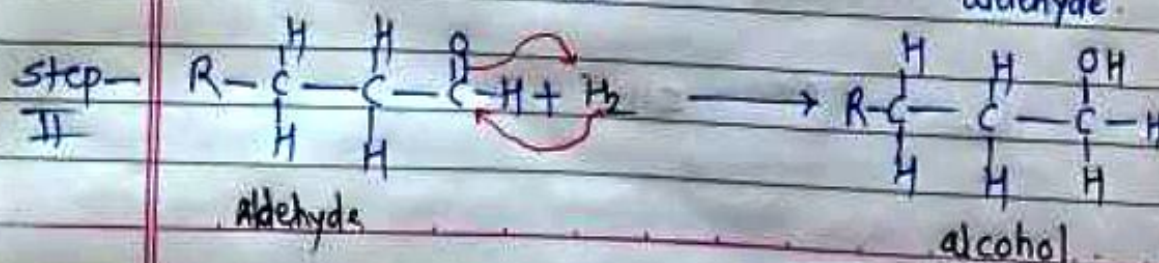
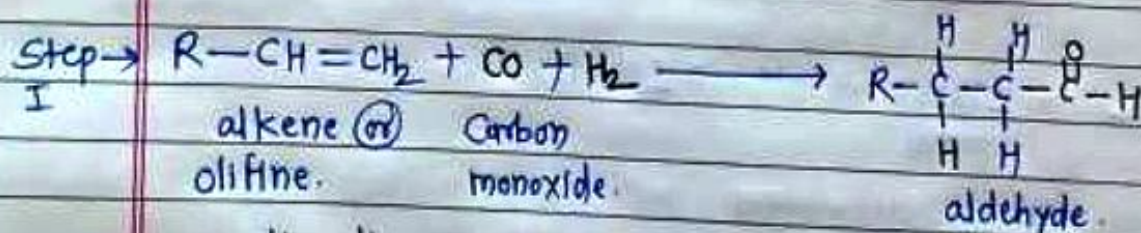
2) Reduction of Unsaturated Compound.



Addⁿ of hydrogen is called as Reduction.

* Hydroformylation of Olefine

Subsequent Reduction of aldehydes into alcohol
olefine = alkene.



Atomic no. filling $\rightarrow [2, 8, 8, 18, 18, 32]$

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* Use of dihydrogen

- 1) Preparation of $\text{NH}_3(\text{g})$ (Ammonia)
- 2) Preparation of $\text{HCl}(\text{g})$ (Hydrochloride)
- 3) Preparation of Vanaspathi ghee
- 4) —|— Making of Rocket fuel.
- 5) —|— of Methanol (CH_3OH).

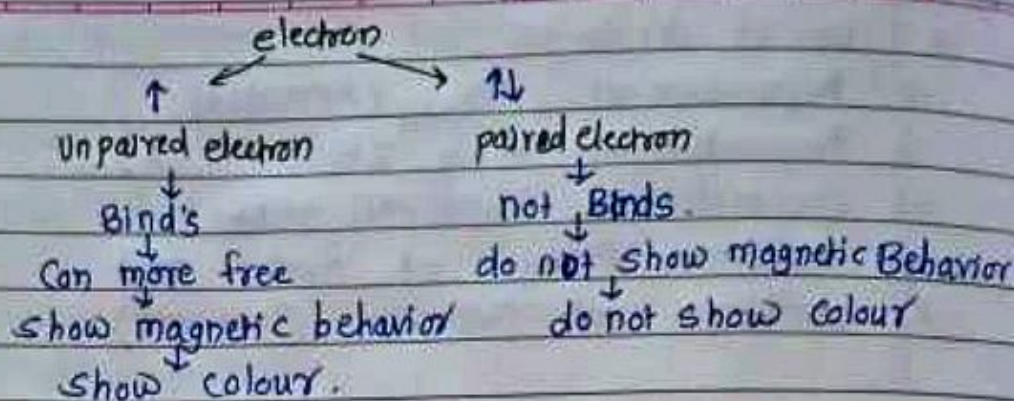
* Element	Alkali metals Li, Na, K, Rb, Cs, Fr	Alkaline earth metals. Be, Mg, Ca, Sr, Ba, Ra
Charge	+1	+2
Group	Group - 1	Group - 2
General e ⁻ Configuration	ns^1	ns^2
Radioactive nature	Fr = Francium	Ra = Radium.

* Electronic Configuration of Group - 1 element

Period	Element	Atomic no.	General electronic Configuration ns^1	Condensed Configuration	Inert gases.
1	H	1	$1s^1$	$1s^1$	[He]
2	Li	3	$1s^2 2s^1$	[He] $2s^1$	[Ne]
3	Na	11	$1s^2 2s^2 2p^6 3s^1$	[Ne] $3s^1$	[Ar]
4	K	19	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$	[Ar] $4s^1$	[Kr]
5	Rb	37	$1s^2 \text{— —} 5s^1$	[Kr] $5s^1$	[Xe]
6	Cs	55	$1s^2 \text{— —} 6s^1$	[Xe] $6s^1$	[Rn]
7	Fr	87	$1s^2 \text{— —} 7s^1$	[Rn] $7s^1$	

Group - 2 element $\rightarrow$ Electronic Configuration.

					[He]
2	Be	4	$1s^2 2s^2$	[He] $2s^2$	[Ne]
3	Mg	12	$1s^2 2s^2 2p^6 3s^2$	[Ne] $3s^2$	[Ar]
4	Ca	20	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$	[Ar] $4s^2$	[Kr]
5	Sr	38	$1s^2 \text{— —} 5s^2$	[Kr] $5s^2$	[Xe]
6	Ba	56	$1s^2 \text{— —} 6s^2$	[Xe] $6s^2$	[Rn]
7	Ra	88	$1s^2 \text{— —} 7s^2$	[Rn] $7s^2$	



* Physical Properties.

	Group - I	Group - II
Colour	Silvery white (Na)	Silvery white, Luster (shining)
General ele ^o Conf:	ns^1	ns^2
Behaviour	$0.5 \Rightarrow +1$ (Unipositive or Univalent)	$\text{Oxidation state} \Rightarrow +2$ (Dipositive or Divalent)
	$\text{Na} \rightarrow ns^1$ $\text{Na}^+ \rightarrow ns^0$ Na^+ confi. is similar to G-18 (stable)	$\text{Mg} \rightarrow ns^2$ $\text{Mg}^{2+} \rightarrow ns^0$ Mg^{2+} confi. is similar to G-18 (stable)
	Magnetic behaviour Dimagnetic	Dimagnetic
	So, it is colourless (Na^+)	So, it is colourless (Mg^{2+})

- As we move down Atomic size increase
- Ionic radii increase & Ionisation energy decreases
- Electronegativity decreases.

$$\text{At. Size} \propto \frac{1}{\text{I.P.}} \propto \frac{1}{\text{EN.}}$$

Problem 8.2 Sodium forms ionic compounds having formulae NaCl , NaH , & Na_2CO_3 . Explain.

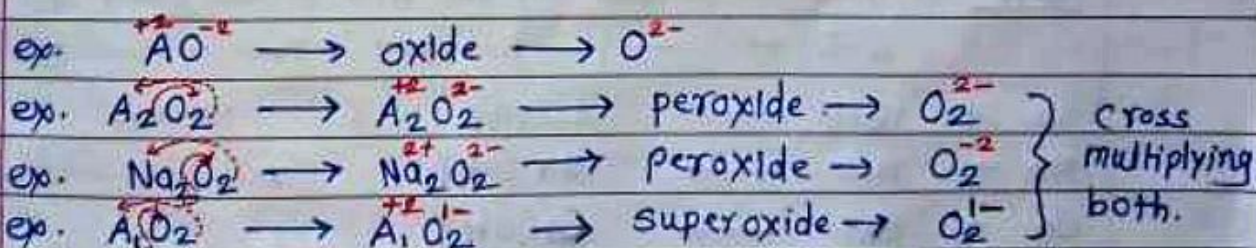
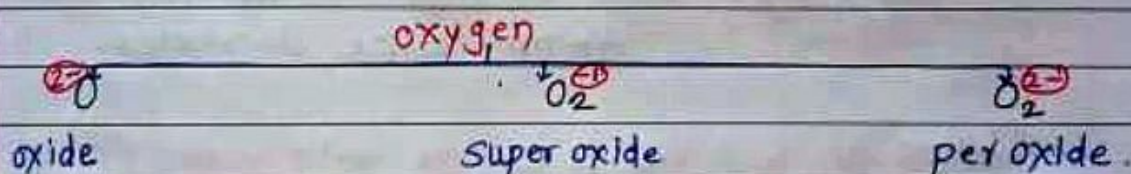
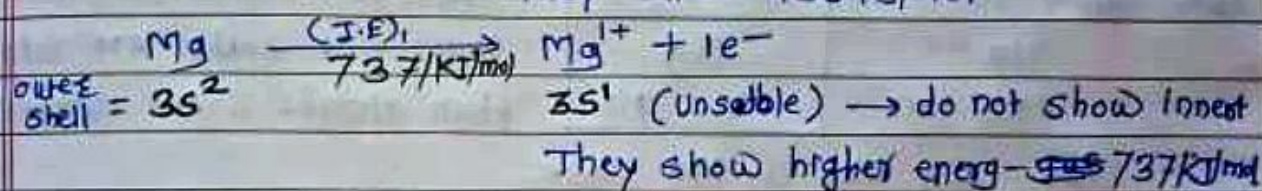
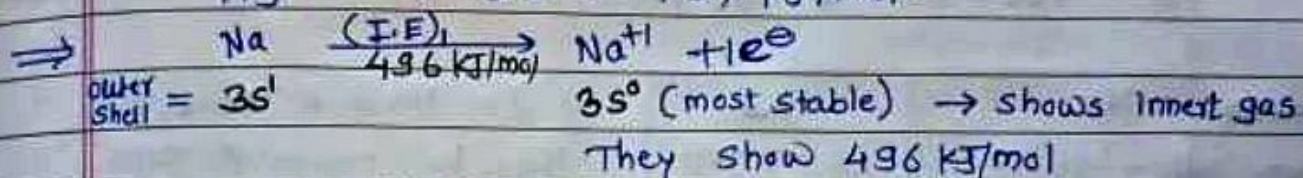
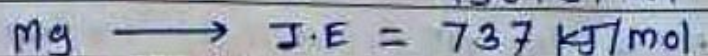


In all cases, Na forms univalent (Na^+) ion.

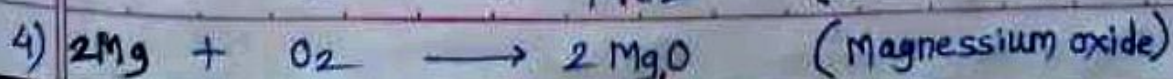


The Configuration similar @ Shows G-18 element than the compound is more stable.

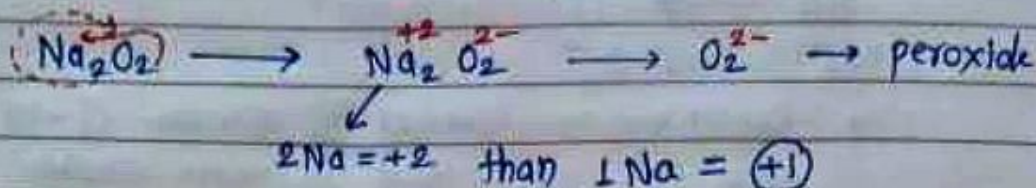
Problem. Explain the observed values 496 kJ/mol & 737 kJ/mol of the first ionization enthalpy of Na & Mg respectively



Reactions of Group-1 & Group-2 with oxygen (Any)



Problem 8.1. What is the oxidation state of sodium (Na) in Na_2O_2 ?



8.5 NOTE $\rightarrow$ G-I $\rightarrow$ Top to bottom $\rightarrow$ size $\rightarrow \uparrow \rightarrow (Z_{\text{eff}}) \rightarrow \downarrow$
In periodic table $\rightarrow$ left to right $\rightarrow$ size $\rightarrow \downarrow \Rightarrow$ effective nuclear charge $\uparrow$ (Z_{eff})

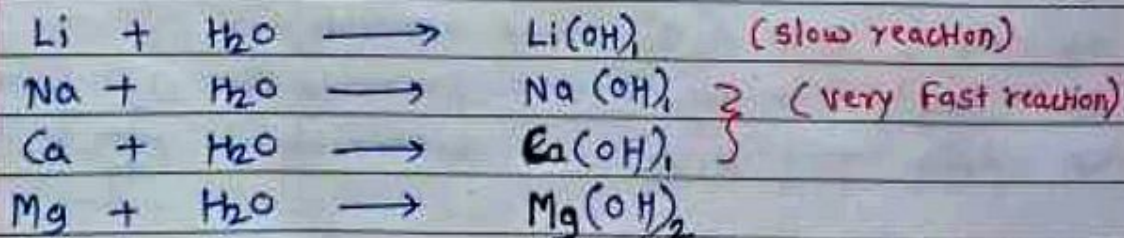
Problem The atomic radii of Na, K & Mg are 186, 227 & 160 pm respectively. Explain the differences

$\Rightarrow$	Na	K	1) In Na & K case,
atomic radii $\rightarrow$	186 pm	227 pm	Na < K, top to Bottom.
	Na	Mg	Number of shells are added.
	186 pm	160 pm	than atomic size increases.

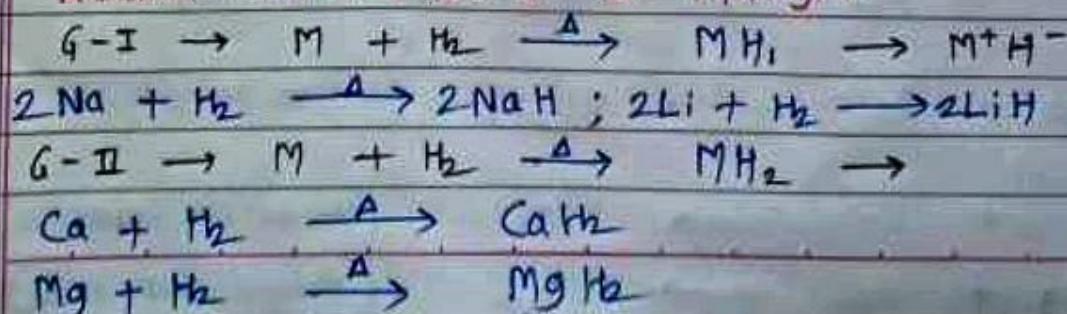
② In Na & Mg case,
Na > Mg. Left to Right Z_{eff} increases than Atomic size decreases.

* Reaction of G-I & G-II element with water (H_2O)

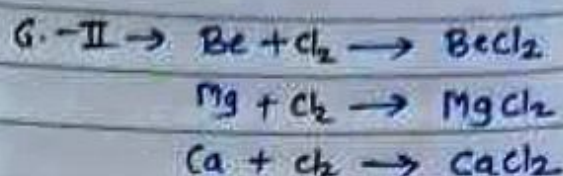
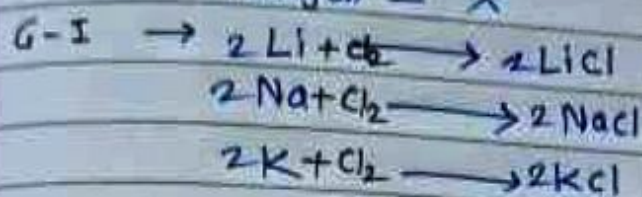
NOTE $\rightarrow$ G-I $\rightarrow$ attached only one atom because charge (+1)
G-II $\rightarrow$ — two atom — — — (+2)



* Reaction with G-I & G-II of Hydrogen.



* Reaction of G-I & G-II with halogen. (Cl, Br, I, F).
Halogen = X



* Reaction of (Mg) G-2 with N₂
 $3\text{Mg} + \text{N}_2 \rightarrow \text{Mg}_3\text{N}_2$

* Reducing Nature (As a Reducing Agent)

G-I

- Alkali metals
- All group-I element show Very high negative electrode potential
- more stronger Reducing Agent
- Li → Is most strongest reducing agent in periodic table.

G-II

- Alkaline earth metal.
- All G-II element show high negative value of electrode potential
- Stronger reducing Agent

Problem 8-6

NaCl is an ionic compound but LiCl has some covalent character, explain.

[NOTE] G-I → Li → size small; Na → size large.
(max. attraction) (min. attraction) @ not

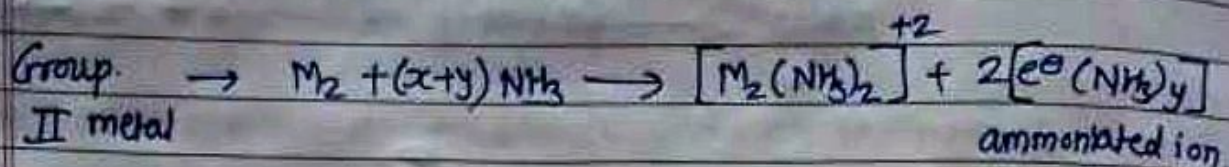
① There for Lithium shows smaller size than maximum attraction to chloride to form covalent bond.

Reason → ② In NaCl, Na⁺ show higher/larger size, Charge density decreases low, chloride do not attract much. So NaCl show Ionic character

- LiCl shows distortion in shape



Relation of Group-I (alkali metals) & Group-II (alkaline earth metals) in solution of liquid ammonia ⁺¹



M-Group-I metal @ Group-2 metal.



* Diagonal Relationship $\rightarrow$

G-I G-II G-III

Points

H

Li Be B

Na Mg Al

K Ca Ga

Rb Sr In

Cs Ba Tl

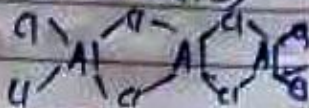
Fo Ra -

The element of one of the group shows similar properties as that of next element of other group & they are placed diagonally & hence the relationship between these two elements, is called as Diagonal relationship.

Diagonal Relationship (similarities between Li & Mg)

Point/Properties	Li (G-I) (+1)	Mg (G-II) (+2)
1) Reaction with air (O_2, N_2)	$Li \xrightarrow{air} Li_2O + Li_3N$	$Mg \xrightarrow{air} MgO + Mg_3N_2$
2) Thermal decomposition of carbonates	$Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2 \uparrow$	$MgCO_3 \xrightarrow{\Delta} MgO + CO_2 \uparrow$
3) Nature of chloride	deliquescent	deliquescent
4) formula of crystalline solid.	$LiCl \cdot 2H_2O$	$MgCl_2 \cdot 8H_2O$

Be & Al.

Point	Be	Al.
1) Nature of Bonding	Covalent	Covalent
2) Group.	G-II $\rightarrow (+2)$	G-13 $\rightarrow (+3)$
3) Structure		
4) Acidic Nature	Strong Lewis acid	Strong Lewis acid.
5) Solubility in organic solvent	Soluble	Soluble
6) Nature	Atrophototoxic	Amphoteric
7) Reaction of oxides with Acids & Bases.	$\text{BeO} + \text{HCl} \rightarrow \text{BeCl}_2 + \text{H}_2\text{O}$ $\text{BeO} + \text{NaOH} \rightarrow \text{Na}_2\text{BeO}_2 + \text{H}_2\text{O}$	$\text{Al}_2\text{O}_3 + \text{HCl} \rightarrow \text{AlCl}_3 + \text{H}_2\text{O}$ $\text{Al}_2\text{O}_3 + \text{NaOH} \rightarrow \text{NaAlO}_2$

* Uses of G-I element

- 1) In Long life batteries, digital watches, calculator
- 2) In Nuclear power station for transforming heat.
- 3) as fertilizer
- 4) for making / manufacturing K_2O_2
- 5) In photoelectric cells.

* Biological importance of G-I element

- 1) Na^+ ion in extracellular fluids, transporting medium.
- 2) Regulates flow of water across membrane
- 3) Participate in transmission of Nerv signal
- 4) Potassium ion in oxidation of glucose

* Uses of G-2 - element

- 1) as Moderators in nuclear reactors.
- 2) Alloy of magnesium & Al is used in Aircrafts
- 3) Calcium ions in growth of bones & teeth
- 4) Barium Sulphate is used in medicines as X-ray
- 5) Radium $\rightarrow$ Radiotherapy for Cancer treatment

* Biological importance of G-2 element

- 1) Mg^{2+} ions as chlorophyll in green plants.

- 2) Mg^{2+} ion in breakage of fat & glucose
- 3) Ca^{2+} ion for bones & teeth.
- 4) Ca^{2+} ion in blood clotting
- 5) Ca^{2+} in stretching & contraction of muscles
- 6) Ca^{2+} in Regular beating of heart.

Problem Magnesium strip slowly tarnishes on keeping in air but metallic calcium is readily attacked by air. Explain.

Mg strip $\xrightarrow{\text{air}}$ slowly React
Ca strip $\xrightarrow{\text{air}}$ React fastly

$\therefore$ attacking nature of metal $\leftarrow$ metallic character $\leftarrow$ tendency to lose electron

→ In case of Mg.

it has less metallic character
less tendency to lose electron, so it react with air.

G-2

Be

Mg

Ca

Sc

Ba

Ra

metallic character increases.

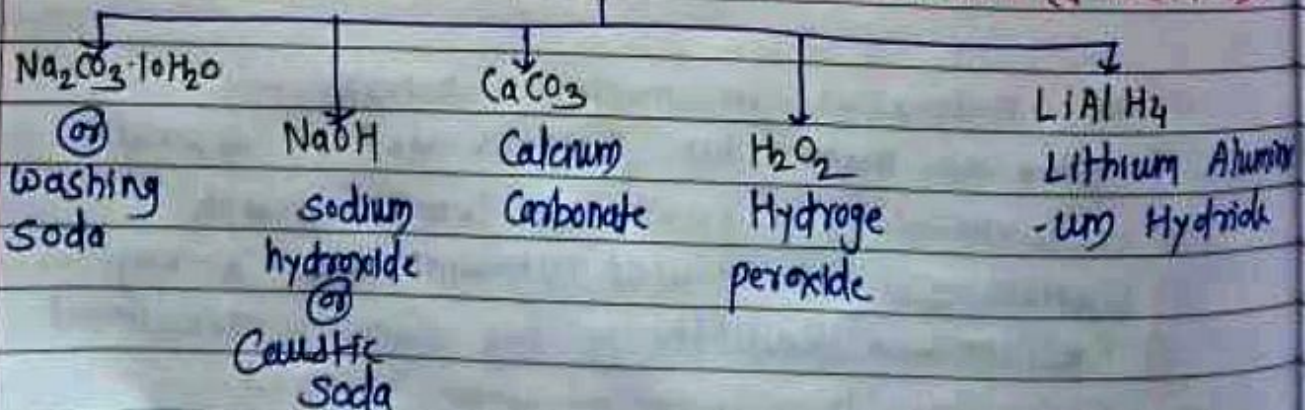


→ In case of Ca,

It has high metallic character
high tendency to lose e^- , so, it react with air fastly



Some important compounds of s-block element (G-I & G-II)



$$(NH_4)HCO_3 + NaCl \longrightarrow NH_4Cl + NaHCO_3$$
$$\text{NaHCO}_3 \xrightarrow{\Delta} \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$$

Sodium carbonate

$$\text{NH}_4\text{Cl} + \text{Ca}(\text{OH})_2 \longrightarrow \text{NH}_3 + \text{CaCl}_2 + \text{H}_2\text{O}.$$

Ammonia.

$$\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O} \xrightarrow{\Delta} \text{Na}_2\text{CO}_3 + 10\text{H}_2\text{O}$$

Unhydrated Sodium Carbonate (dry)

$$\text{Na}_2\text{CO}_3 + \text{H}_2\text{O} \longrightarrow \text{NaHCO}_3 + \text{NaOH}$$

Sodium bicarbonate.

Solution nature is alkaline (Base)

i) Used as cleaning material

2) Hard water $\longrightarrow$ soft water

$$\text{Ca}(\text{HCO}_3)_2 + \text{Na}_2\text{CO}_3 \longrightarrow \text{CaCO}_3 + \text{NaHCO}_3$$

3) production of soap & caustic soda

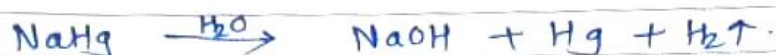
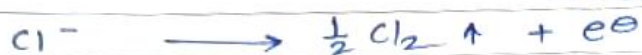
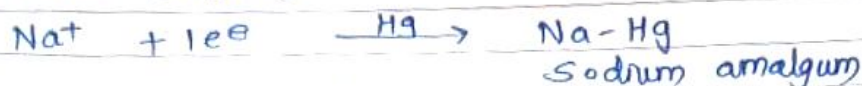
4) Used as Laboratory Reagent.

② NaOH (or) Caustic soda

Preparation of NaOH



Brine solution



* Properties of NaOH

NaOH $\longrightarrow$ nature $\longrightarrow$ deliquescent
(tendency to absorb moisture)

- Soluble in H_2O
- absorb moisture
- NaOH absorb $CO_2(g)$



* Uses

- Purification of Bauxite (Al-ore)
- Production of soap, paper, silk, chemicals
- Petroleum refining
- as Laboratory Reagent.

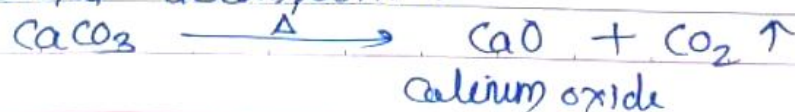
③ $CaCO_3$ (or) Calcium Carbonate

I) Method of Preparation



* Properties.

i) Thermal decomposition



2) Reaction with acid (HCl)



3) Reaction with H_2SO_4

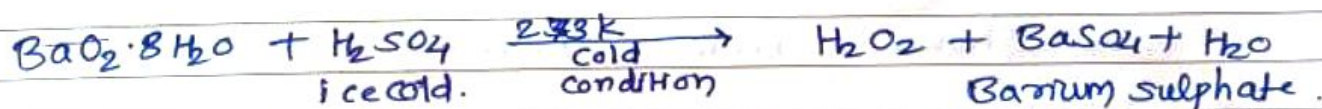


Uses.

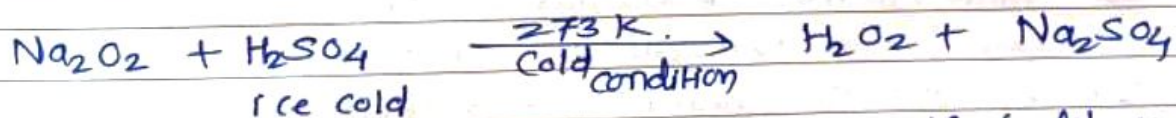
- As Marble in Building material.
- In Cement manufacturing
- In extraction of metals from Ore.
- In toothpaste, chewing gum.
- High quality papers.

4) H_2O_2 (or) Hydrogen peroxide.

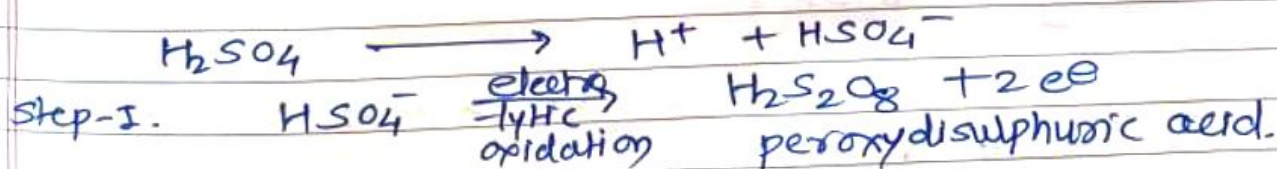
1) $\text{BaO}_2 \cdot 8\text{H}_2\text{O}$ (Hydrated Barium peroxide)



2) Na_2O_2 (or) sodium peroxide.



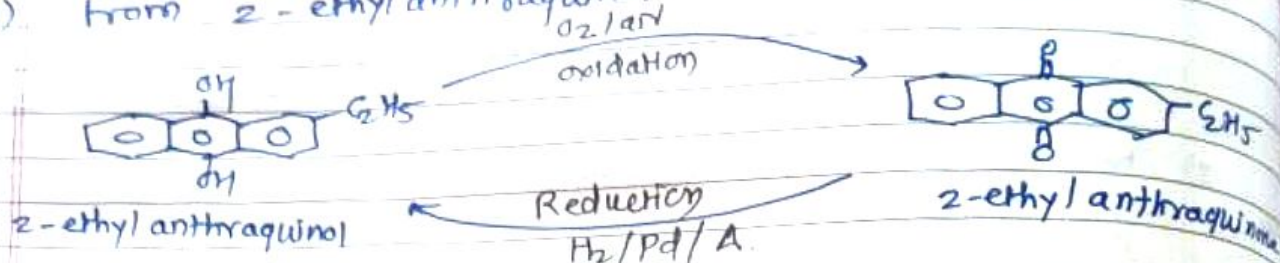
3) Electrolytic oxidation of 50% aq. solⁿ of dil. H_2SO_4 .



Step-II



4) from 2-ethylanthraquinol.



* Properties of H_2O_2

- Pure $\text{H}_2\text{O}_2 \rightarrow$ pale blue coloured liquid.
- H_2O_2 is miscible in H_2O (water)
- $\text{H}_2\text{O}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{O}_2 \cdot \text{H}_2\text{O}$
- Strength of aq. solution of $\text{H}_2\text{O}_2 \rightarrow$ measured in term of volume.
- 30% aq. solution of $\text{H}_2\text{O}_2 =$ strength-volume of 100
- H_2O_2 acts as oxidizing & Reducing

(a)

H_2O_2 shows oxidizing & Reducing action.

* Uses

- Mouth washing, antiseptic
- as milk, wine, bleaching agent preservative.
- antichlor $\rightarrow$ Removal of Chlorine.
- In environment Chemistry $\rightarrow$ Pollution controller
- & also for Restoration of aerobic condition to Sewage water.

Problem
8.8

Calculate % (by mass) of a H_2O_2 solution which is 45.4 Volume.

% mass of H_2O_2 solution = strength = g/ltr.
Volume = 45.4



$$\begin{aligned} \text{molar mass} &= 2 \times (2 \times 1 + 2 \times 16) \\ &= 2(2 + 32) \\ &= 2 \times 34 = 68 \end{aligned}$$

$$V_m = 22.4.$$

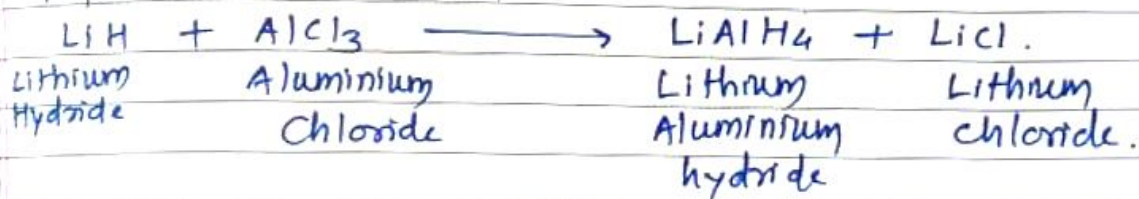
$$\begin{aligned} \text{mass } 68 &= 22.4 \quad \leftarrow \text{volume} \\ (x) &= 45.4 \quad \leftarrow \text{volume.} \end{aligned}$$

$$x = \frac{68 \times 45.4}{22.4} = 136 \rightarrow \text{mass}$$

$$\text{strength} = \% \text{ mass in solution} = \frac{136}{1000} = 13.6 \% \text{ mass.}$$

LiAlH_4 (Lithium Aluminium Hydride)

Preparation of LiAlH_4



2) Chemical Reaction LiAlH_4

a) Source of Hydride (H^-) $\rightarrow$ preparation of phosphine.



2) As a Reducing Agent (H_2 addition)

